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*Title of the Project*  
*Automated Document Scanner & OCR*  
*Enhancement System*

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*Course Name: Computer Vision*  
*(CSEN4236)*

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*Date of Submission: 15 April 2026*

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# ABSTRACT

In the modern digital era, efficient document digitization plays a crucial role in data storage, retrieval, and automation. This project presents an automated document scanner and Optical Character Recognition (OCR) enhancement system using computer vision techniques. The system detects document boundaries using edge detection and contour analysis, followed by perspective transformation to obtain a top-down scanned view of the document.

To enhance readability, preprocessing techniques such as grayscale conversion, adaptive thresholding, and contrast enhancement are applied. These techniques reduce noise and improve text clarity, enabling accurate OCR. The system integrates Tesseract OCR to extract textual content from processed images.

The proposed solution offers a cost-effective alternative to traditional scanning systems and can handle variations in lighting, orientation, and background conditions. The implementation uses Python, OpenCV, and Tesseract OCR.

The expected outcome is a robust document digitization system capable of extracting readable text from images. The system has applications in digital archiving, automated documentation, and smart scanning applications.

**Keywords: OCR, OpenCV, Image Processing, Tesseract, Document Scanner**

## 3. INTRODUCTION

### 3.1 Motivation

With the rapid growth of digital systems, converting physical documents into digital format has become essential. Manual scanning methods are time-consuming and inefficient. This project aims to automate document detection and text extraction using computer vision techniques.

### 3.2 Contribution

- Automated document boundary detection
- Perspective correction using transformation
- Multiple preprocessing techniques for enhancement
- OCR-based text extraction using Tesseract
- Web-based interface for easy usage

## 4. LITERATURE REVIEW

### Existing Research Work

- Tesseract OCR engine for text recognition
- OpenCV-based document detection systems
- Mobile scanning applications like CamScanner

### Gap Analysis

- Existing systems struggle with noisy backgrounds
- Limited preprocessing reduces OCR accuracy
- Lack of automation in traditional approaches

### Proposed Approach

- Multi-step preprocessing pipeline
- Improved document detection
- Better OCR accuracy through enhancement

# 5. PROPOSED METHODOLOGY

## 5.1 Model Description

The proposed system is designed as a multi-stage pipeline that combines computer vision techniques and optical character recognition to automate document scanning and text extraction. The implementation is carried out using Python, with OpenCV as the primary library for image processing and Tesseract OCR for text recognition.

The system takes an input image containing a document and processes it through several stages including preprocessing, document boundary detection, geometric transformation, image enhancement, and text extraction. Each stage contributes to improving the accuracy and readability of the final output.

OpenCV is used extensively for tasks such as edge detection, contour identification, and perspective transformation. Tesseract OCR is integrated to extract textual information from the processed image. Multiple preprocessing techniques such as adaptive thresholding, CLAHE (Contrast Limited Adaptive Histogram Equalization), and denoising are applied to improve OCR performance under varying conditions.

The system is designed to handle variations in lighting, orientation, and background noise, making it robust for real-world applications such as document digitization and automation.

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## 5.2 Algorithm / Processing Pipeline

The complete workflow of the system is described below:

### Step 1: Image Acquisition (Input Stage)

The system accepts an image input either from a file upload or a camera capture. The input image may contain a document placed on a surface with possible distortions such as rotation, perspective skew, or uneven lighting.

### Step 2: Preprocessing and Grayscale Conversion

The input image is converted into grayscale to simplify further processing. This reduces computational complexity and removes unnecessary color information. A Gaussian blur is applied to smooth the image and reduce noise, which helps in improving edge detection accuracy.

### **Step 3: Edge Detection (Canny Edge Detector)**

The Canny edge detection algorithm is applied to identify the edges in the image. This step highlights the boundaries of objects, particularly the document edges, by detecting intensity gradients in the image.

### **Step 4: Contour Detection and Selection**

Contours are extracted from the edge-detected image. These contours represent continuous boundaries of shapes. The system identifies the largest contours and approximates them into polygonal curves. The contour with four points (quadrilateral shape) is selected as the document boundary.

### **Step 5: Perspective Transformation (Document Rectification)**

Once the document boundary is detected, a four-point perspective transformation is applied. This step converts the skewed document into a top-down view, similar to a scanned image. It corrects distortions caused by camera angle and alignment.

### **Step 6: Image Enhancement for OCR**

The transformed image is further processed to enhance text visibility. Various preprocessing techniques are applied, including:

- Adaptive Thresholding (for varying lighting conditions)
- Otsu Thresholding (for global thresholding)
- CLAHE (to improve contrast)
- Morphological Operations (to reduce noise)
- Denoising filters

These techniques improve the contrast between text and background, making it easier for OCR to detect characters.

### **Step 7: Optical Character Recognition (OCR)**

The enhanced image is passed to the Tesseract OCR engine. Tesseract analyzes the image and converts the visual text into machine-readable text. The extracted text is then displayed to the user.

## Step 8: Output Generation

The final output includes:

- Scanned document (cleaned image)
- Enhanced image for OCR
- Extracted text

The system may also allow users to download the processed document or copy the extracted text.

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## 5.3 System Workflow Summary

The complete workflow can be summarized as:

Input Image → Preprocessing → Edge Detection → Contour Detection  
→ Perspective Transformation → Image Enhancement → OCR → Output Text

# 6. RESULTS AND DISCUSSION

- The system successfully detects document boundaries
- Perspective transformation produces a scanned effect
- OCR performs well on printed text
- Performance reduces with noisy backgrounds and handwritten text

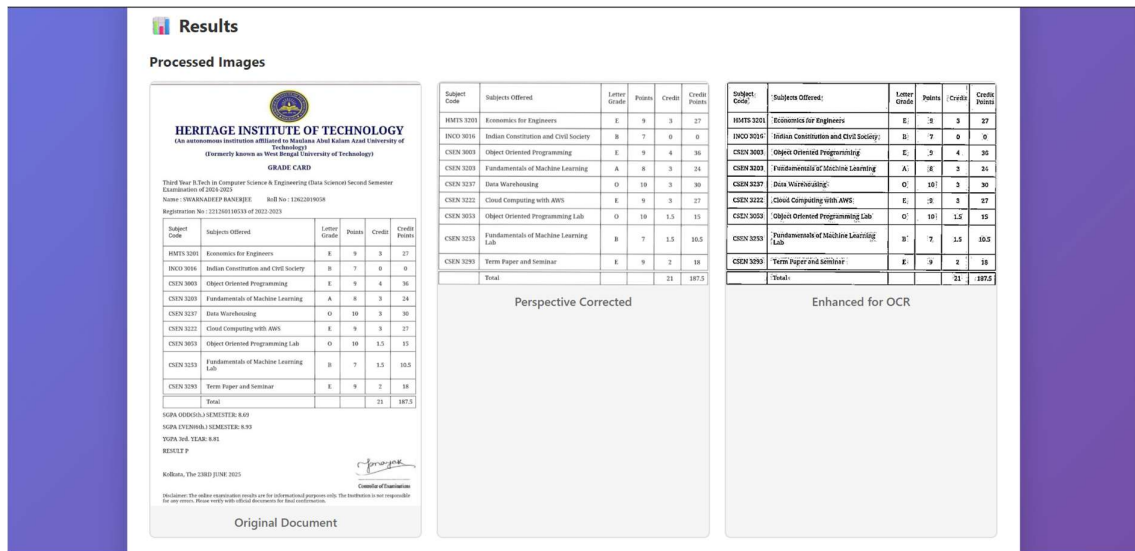
The screenshot displays a web application interface for document digitization. The header reads "Advanced document digitization with text extraction". The interface is divided into two main sections: "Upload Document" and "How It Works".

**Upload Document Section:**

- A tip box states: "Tip: Upload a document image for automatic scanning and text extraction".
- A "Select Image File" section shows a file named "cloud computing.jpg" with a dashed border and a small image icon.
- Below the file selection, it says "Supported: JPG, PNG, BMP, TIFF (Max 10MB)".
- A "Preprocessing Method" dropdown menu is set to "Adaptive Thresholding (Recommended)".
- An "OCR Language" dropdown menu is set to "English".
- A blue button at the bottom is labeled "Scan Document".

**How It Works Section:**

- 1 Document Detection:** Edge detection and contour analysis to identify document boundaries.
- 2 Perspective Correction:** Four-point transformation to correct skewed documents.
- 3 Image Enhancement:** Advanced preprocessing techniques for optimal OCR performance.
- 4 Text Extraction:** Tesseract OCR with multi-language support.
- Preprocessing Methods:**
  - **Adaptive:** Best for varying lighting conditions
  - **Otsu:** Automatic global thresholding
  - **CLAHE:** Contrast enhancement
  - **Morphological:** Noise reduction



## 7. CONCLUSION AND FUTURE SCOPE

### Conclusion

The project successfully demonstrates an automated document scanning and OCR system using computer vision techniques. It reduces manual effort and improves efficiency in document digitization.

### Future Scope

- Improved handwritten text recognition
- Integration with mobile applications
- Cloud-based document storage
- AI-based noise removal

## 8. REFERENCES

1. R. Smith, "An Overview of the Tesseract OCR Engine"
2. OpenCV Documentation – <https://opencv.org>
3. Python Documentation – <https://docs.python.org>
4. Tesseract OCR Documentation – <https://github.com/tesseract-ocr>

localhost:5000

### Results

(An autonomous institution affiliated to Madrasa Abdul Kalam University of Technology)  
(Formerly known as West Bengal University of Technology)

**GRADE CARD**

Third Year B.Tech In Computer Science & Engineering (Old Scheme) Second Semester  
Examinations of 2023-2025  
Name : HANSHABANDHU BANERJEE    Roll No : ENG23009008  
Registration No : 12328018013 of 2023-2025

| Subject Code  | Subjects Offered                      | Letter Grade | Points | Credits | Credit Points      |
|---------------|---------------------------------------|--------------|--------|---------|--------------------|
| HNMTS 3001    | Economics for Engineers               | E            | 9      | 3       | 27                 |
| INDO 3014     | Indian Constitution and Civil Society | B            | 7      | 0       | 0                  |
| CSSE 3003     | Object Oriented Programming           | E            | 9      | 4       | 36                 |
| CSSE 3009     | Fundamentals of Machine Learning      | A            | 8      | 3       | 24                 |
| CSSE 3237     | Data Warehousing                      | O            | 10     | 3       | 30                 |
| CSSE 3232     | Cloud Computing with AWS              | E            | 9      | 3       | 27                 |
| CSSE 3013     | Object Oriented Programming Lab       | O            | 10     | 1.5     | 15                 |
| CSSE 3233     | Fundamentals of Machine Learning Lab  | B            | 7      | 1.5     | 10.5               |
| CSSE 3203     | Term Paper and Seminar                | E            | 9      | 2       | 18                 |
| <b>Total:</b> |                                       |              |        |         | <b>21    187.5</b> |

SGPA: 000016161 SEMESTER: B.09

SGPA: 021000161 SEMESTER: B.03

SGPA: 046.1000: B.01

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**Original Document**

**Processed Images**

**Perspective Corrected**

**Enhanced for OCR**



# Document Scanner & OCR

Advanced document digitization with text extraction

## Upload Document

**Tip:** Upload a document image for automatic scanning and text extraction

Select Image File

test2.jpeg

Supported: JPG, PNG, BMP, TIFF (Max 16MB)

Preprocessing Method

Adaptive Thresholding (Recommended)

Adaptive Thresholding (Recommended)

Otsu's Thresholding

CLAHE Enhancement

Morphological Operations

Denoising

Scan Document

## How It Works

### 1 Document Detection

Edge detection and contour analysis to identify document boundaries

### 2 Perspective Correction

Four-point transformation to correct skewed documents

### 3 Image Enhancement

Advanced preprocessing techniques for optimal OCR performance

### 4 Text Extraction

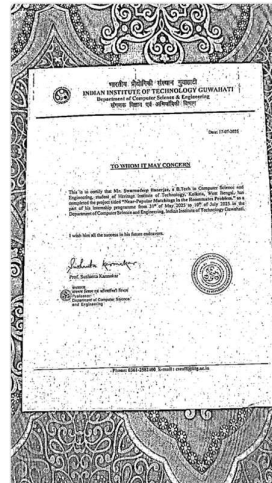
Tesseract OCR with multi-language support

### Preprocessing Methods

- Adaptive: Best for varying lighting conditions
- Otsu: Automatic global thresholding
- CLAHE: Contrast enhancement

## Results

### Processed Images



## Extracted Text

(No text detected)

Copy Text

Download

Scan Another